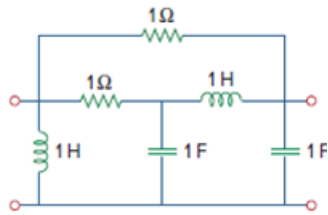


Problem

At $\omega = 1$ rad/s, find the transmission parameters of the network in Fig. 19.127 using *PSpice* or *MultiSim*.

Figure 19.127



Step-by-step solution

Step 1 of 18

Consider the following circuit diagram:

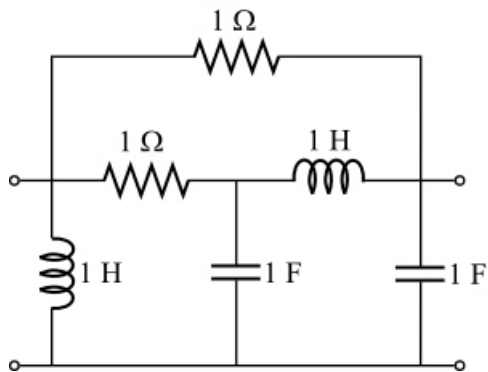


Figure 1

Step 2 of 18

Theoretical analysis:

The value of the angular frequency, $\omega = 1$ rad/s.

Calculate the impedance value of inductor.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(1)(1) \\ &= j\Omega \end{aligned}$$

Calculate the impedance value of capacitor.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(1)(1)} \\ &= -j\Omega \end{aligned}$$

Step 3 of 18

Draw the frequency domain circuit.

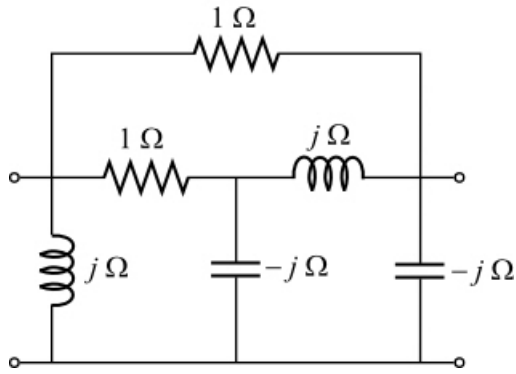


Figure 2

Step 4 of 18

Write transition parameters:

$$\mathbf{A} = \left. \frac{\mathbf{V}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0} \dots\dots (1)$$

$$\mathbf{B} = - \left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \dots\dots (2)$$

$$\mathbf{C} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0} \dots\dots (3)$$

$$\mathbf{D} = - \left. \frac{\mathbf{I}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \dots\dots (4)$$

To calculate the values of **A** and **C** parameters, connect a test voltage source to input port and open circuit the output port. Draw the modified circuit diagram.

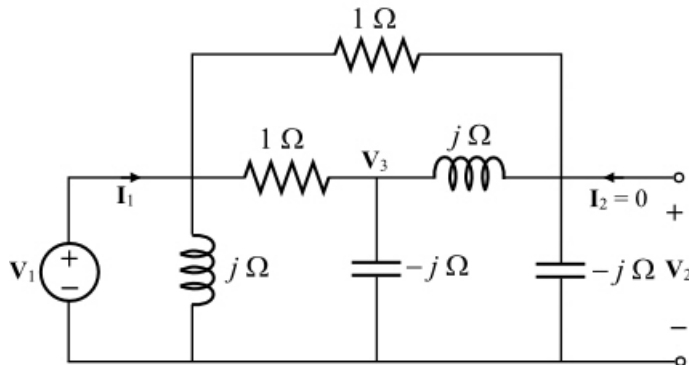


Figure 3

Apply Kirchhoff's current law at node $\mathbf{V}_2$.

$$\begin{aligned} \frac{\mathbf{V}_2 - \mathbf{V}_3}{j} + \frac{\mathbf{V}_2}{-j} + \frac{\mathbf{V}_2 - \mathbf{V}_1}{1} &= 0 \\ -\mathbf{V}_1 + \left(\frac{1}{j} - \frac{1}{j} + 1 \right) \mathbf{V}_2 - \frac{\mathbf{V}_3}{j} &= 0 \\ -\mathbf{V}_1 + \mathbf{V}_2 + j\mathbf{V}_3 &= 0 \\ j\mathbf{V}_3 &= \mathbf{V}_1 - \mathbf{V}_2 \\ \mathbf{V}_3 &= j(\mathbf{V}_2 - \mathbf{V}_1) \dots\dots (5) \end{aligned}$$

Step 5 of 18

Apply Kirchhoff's current law at node V_3 .

$$\frac{V_3 - V_2}{j} + \frac{V_3}{-j} + \frac{V_3 - V_1}{1} = 0$$

$$-V_1 - \frac{V_2}{j} + \left(\frac{1}{j} - \frac{1}{j} + 1\right)V_3 = 0$$

$$-V_1 + jV_2 + V_3 = 0$$

Substitute $j(V_2 - V_1)$ for V_3 in the above equation.

$$-V_1 + jV_2 + j(V_2 - V_1) = 0$$

$$(-1 - j)V_1 + j2V_2 = 0$$

$$(1 + j)V_1 = j2V_2$$

$$V_1 = \left(\frac{j2}{1 + j}\right)V_2$$

$$V_1 = (1 + j)V_2 \dots\dots (6)$$

Step 6 of 18

Apply Kirchhoff's current law at node V_1 .

$$I_1 = \frac{V_1 - V_2}{1} + \frac{V_1}{j} + \frac{V_1 - V_3}{1}$$

$$= \left(1 + \frac{1}{j} + 1\right)V_1 - V_2 - V_3$$

$$= (2 - j)V_1 - V_2 - V_3$$

Substitute $j(V_2 - V_1)$ for V_3 in the above equation.

$$I_1 = (2 - j)V_1 - V_2 - j(V_2 - V_1)$$

$$= (2 - j + j)V_1 - (1 + j)V_2$$

$$= 2V_1 - (1 + j)V_2$$

Now substitute $(1 + j)V_2$ for V_1 in the above equation.

$$I_1 = 2(1 + j)V_2 - (1 + j)V_2$$

$$= (2 + 2j - 1 - j)V_2$$

$$= (1 + j)V_2$$

Step 7 of 18

Substitute $(1+j)V_2$ for V_1 in the equation (1).

$$\begin{aligned} A &= \frac{V_1}{V_2} \\ &= \frac{(1+j)V_2}{V_2} \\ &= 1+j \\ &= 1.414\angle 45^\circ \end{aligned}$$

Substitute $(1+j)V_2$ for I_1 in the equation (2).

$$\begin{aligned} C &= \frac{I_1}{V_2} \\ &= \frac{(1+j)V_2}{V_2} \\ &= 1+j \\ &= 1.414\angle 45^\circ S \end{aligned}$$

Step 8 of 18

To calculate the values of **B** and **D** parameters, connect a test voltage source to input port and short circuit the output port. Draw the modified circuit diagram.

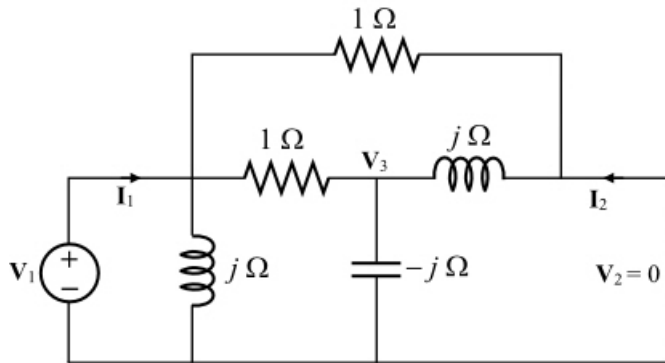


Figure 4

Apply Kirchhoff's current law at node V_2 .

$$\begin{aligned} I_2 &= \frac{-V_3}{j} + \frac{-V_1}{1} \\ &= -V_1 + jV_3 \dots\dots (7) \end{aligned}$$

Apply Kirchhoff's current law at node V_3 .

$$\begin{aligned} \frac{V_3}{j} + \frac{V_3}{-j} + \frac{V_3 - V_1}{1} &= 0 \\ -V_1 + \left(\frac{1}{j} - \frac{1}{j} + 1 \right) V_3 &= 0 \\ -V_1 + V_3 &= 0 \\ V_3 &= V_1 \end{aligned}$$

Step 9 of 18

Substitute $\mathbf{V}_1$ for $\mathbf{V}_3$ in the equation (7).

$$\begin{aligned}\mathbf{I}_2 &= -\mathbf{V}_1 + j\mathbf{V}_1 \\ &= (-1+j)\mathbf{V}_1\end{aligned}$$

Apply Kirchhoff's current law at node $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{I}_1 &= \frac{\mathbf{V}_1}{1} + \frac{\mathbf{V}_1}{j} + \frac{\mathbf{V}_1 - \mathbf{V}_3}{1} \\ &= \left(1 + \frac{1}{j} + 1\right)\mathbf{V}_1 - \mathbf{V}_3 \\ &= \left(1 + \frac{1}{j} + 1\right)\mathbf{V}_1 - \mathbf{V}_1 \quad (\text{since } \mathbf{V}_3 = \mathbf{V}_1) \\ &= (1-j)\mathbf{V}_1\end{aligned}$$

Step 10 of 18

Substitute $(-1+j)\mathbf{V}_1$ for $\mathbf{I}_2$ in the equation (2).

$$\begin{aligned}\mathbf{B} &= -\frac{\mathbf{V}_1}{\mathbf{I}_2} \\ &= -\frac{\mathbf{V}_1}{(-1+j)\mathbf{V}_1} \\ &= 0.5 + j0.5 \\ &= 0.707\angle 45^\circ\end{aligned}$$

Substitute $(-1+j)\mathbf{V}_1$ for $\mathbf{I}_2$ and $(1-j)\mathbf{V}_1$ for $\mathbf{I}_1$ in the equation (4).

$$\begin{aligned}\mathbf{D} &= -\frac{\mathbf{I}_1}{\mathbf{I}_2} \\ &= -\frac{(1-j)\mathbf{V}_1}{(-1+j)\mathbf{V}_1} \\ &= 1\end{aligned}$$

Step 11 of 18

Therefore, the transmission parameters, $\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}$ is $\begin{bmatrix} 1.414\angle 45^\circ & 0.707\angle 45^\circ \\ 1.414\angle 45^\circ & 1 \end{bmatrix}$.

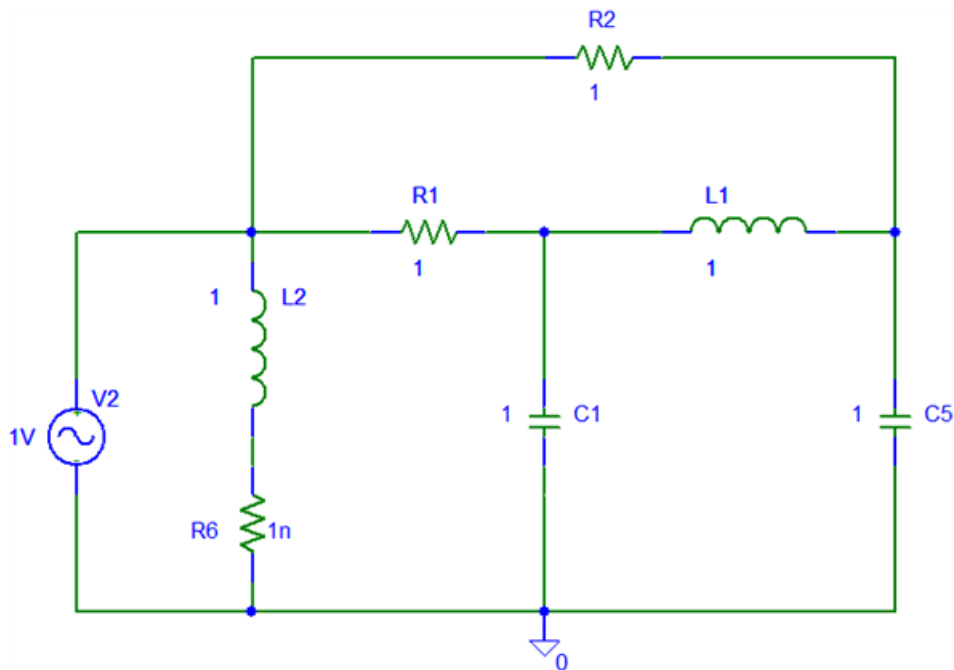
Step 12 of 18

PSpice analysis:

- Open the CIS Demo software.
- Once you are in the capture session frame, click on the menu item File, select new, and then click on Project.
- In the New Project box, type 'selected file name' in the Name text box. Ensure that the Analog or Mixed-signal circuit Wizard is activated.
- Connected the circuit diagram as per the Circuit.
- Double click on each component and enter the given values.
- Click the New Simulation Profile icon and enter a name in the Name text box. You will need to enter the appropriate settings for this project in the Simulation
- Setting box. Click the **Analysis tab**, and select **AC sweep** from the Analysis type list. Then select linear and enter total points as 1 and starting and ending frequency as 0.159155. Click OK and save the document.
- Click on the run icon. You will observe the A/D Demo screen. When you close this screen, you will observe the voltages and currents.

To calculate the values of **A** and **C** parameters, connect a test voltage source to input port and open circuit the output port.

The following is the PSpice schematic circuit:



Connect a IPRINT in series with voltage source and connect a VPRINT1 across the capacitor. Modified circuit is shown below.

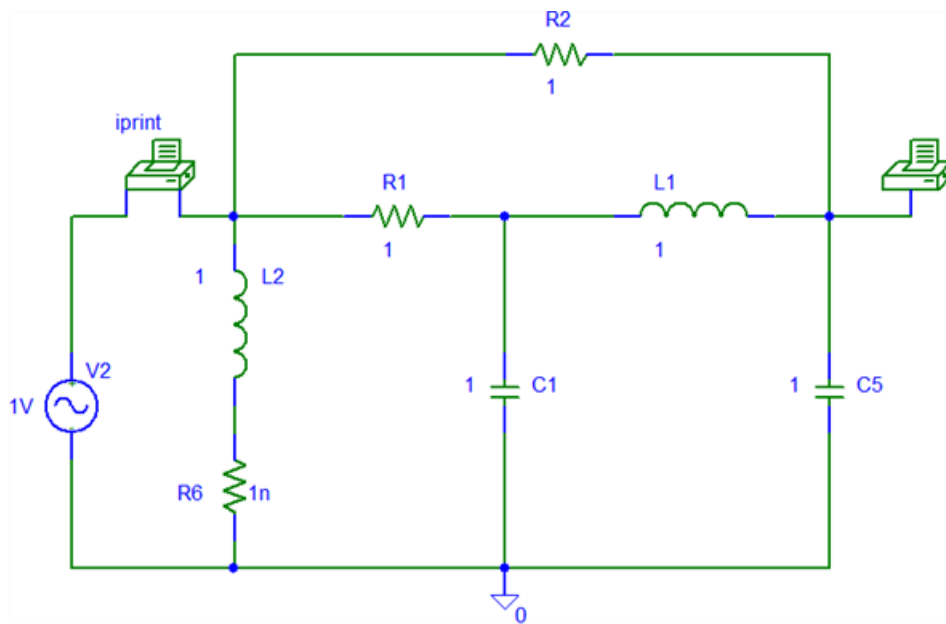


Figure 4

Step 14 of 18

Run the PSpice schematic the following are the obtained current and voltage values.

FREQ IM(V_PRINT11) IP(V_PRINT11)

1.592E-01 1.000E+00 2.049E-05

FREQ VM(\$N_0004) VP(\$N_0004)

1.592E-01 7.071E-01 -4.500E+01

Step 15 of 18

Substitute $1\angle 0^\circ$ for V_1 and $0.707\angle -45^\circ$ for V_1 in the equation (1).

$$\begin{aligned} A &= \frac{V_1}{V_2} \\ &= \frac{1}{0.707\angle -45^\circ} \\ &= 1.414\angle 45^\circ \end{aligned}$$

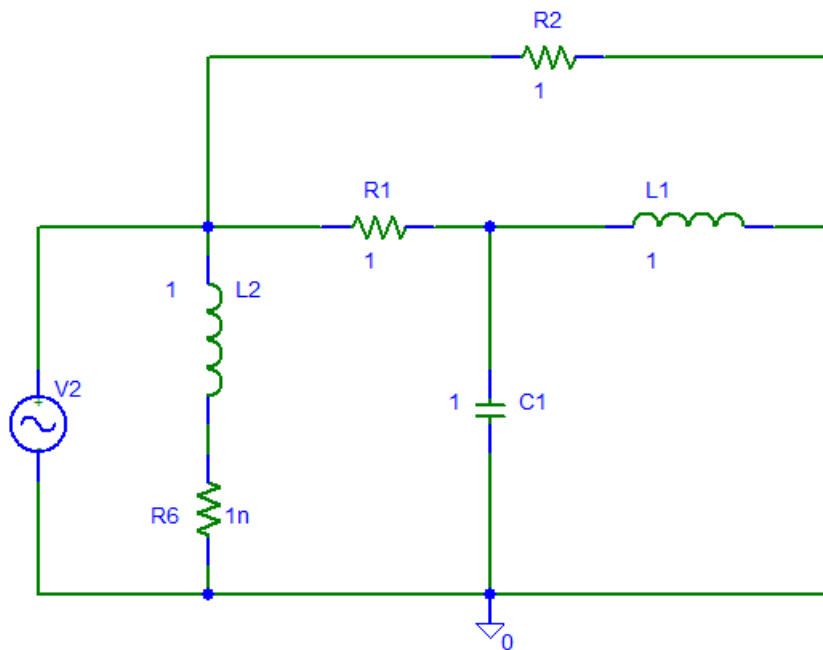
Substitute 1 for I_1 and $0.707\angle -45^\circ$ for V_1 the equation (2).

$$\begin{aligned} C &= \frac{I_1}{V_2} \\ &= \frac{1}{0.707\angle -45^\circ} \\ &= 1.414\angle 45^\circ \text{ S} \end{aligned}$$

Step 16 of 18

To calculate the values of **B** and **D** parameters, connect a test voltage source to input port and short circuit the output port.

The following is the PSpice schematic circuit:



Step 17 of 18

Connect a IPRINT in series with voltage source and connect another IPRINT through short circuit path. Modified circuit is shown below.

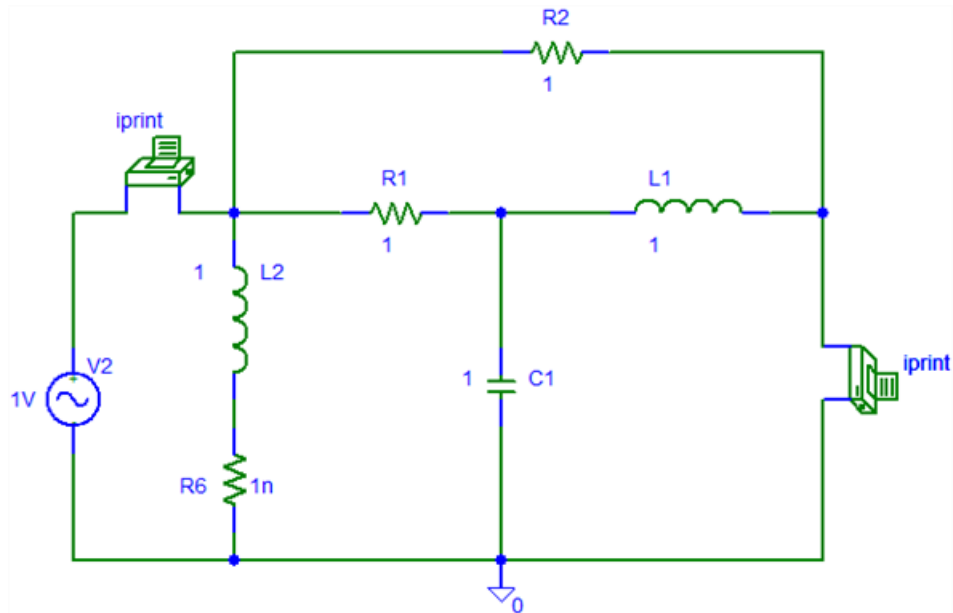


Figure 4

Run the PSpice schematic the following are the obtained current and voltage values.

FREQ IM(V_PRINT11) IP(V_PRINT11)

1.592E-01 1.414E+00 -4.500E+01

FREQ IM(V_PRINT12) IP(V_PRINT12)

1.592E-01 1.414E+00 1.350E+02

Step 18 of 18

Substitute 1 for V_1 and $1.414\angle 135^\circ$ for I_2 in the equation (2):

$$\begin{aligned} B &= -\frac{V_1}{I_2} \\ &= -\frac{1}{1.414\angle 135^\circ} \\ &= 0.707\angle 45^\circ \end{aligned}$$

Substitute $1.414\angle -45^\circ$ for I_1 and $1.414\angle 135^\circ$ for I_2 in the equation (4).

$$\begin{aligned} D &= -\frac{I_1}{I_2} \\ &= -\frac{1.414\angle -45^\circ}{1.414\angle 135^\circ} \\ &= 1 \end{aligned}$$

Therefore, the transmission parameters, $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is $\begin{bmatrix} 1.414\angle 45^\circ & 0.707\angle 45^\circ \\ 1.414\angle 45^\circ & 1 \end{bmatrix}$.